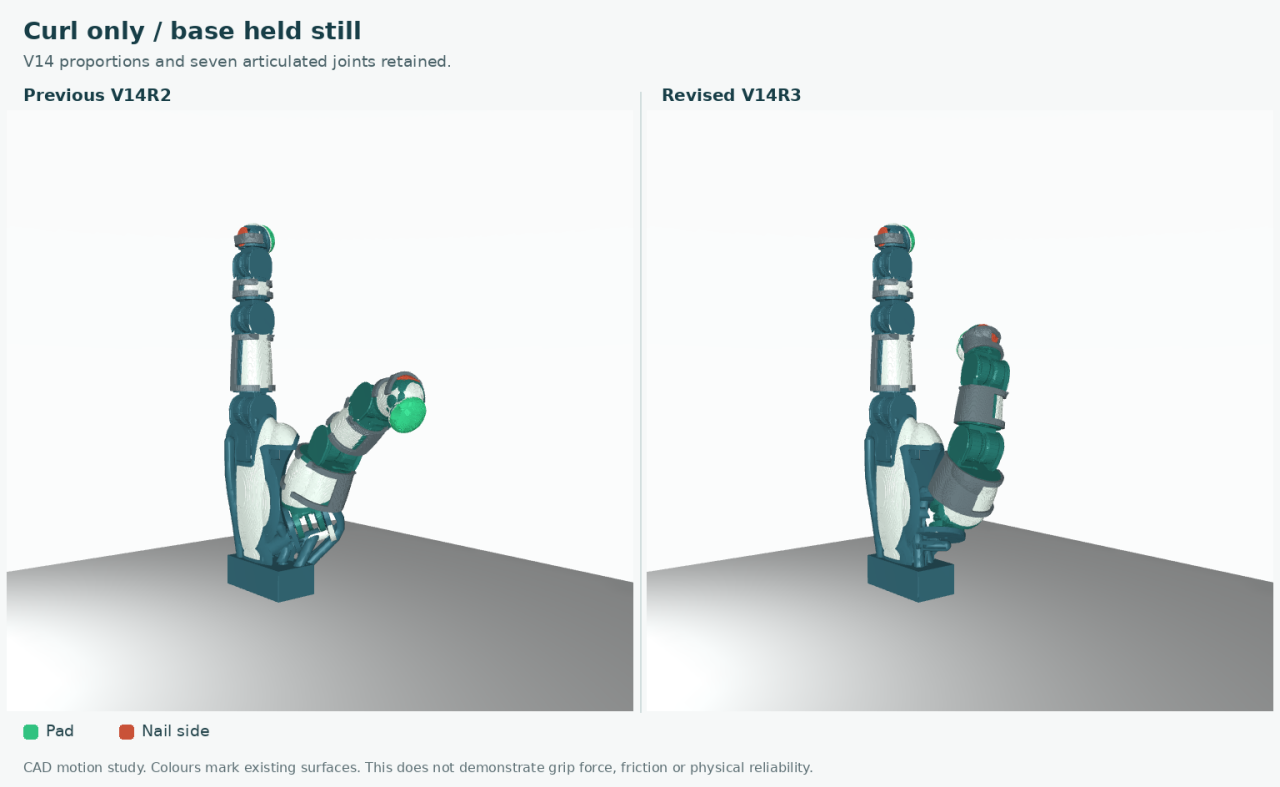
The thumb was facing the wrong way.

V14R3 corrects the thumb orientation and opposition axis while retaining the V14 hand architecture.

The imported rest frame pointed the thumb pulp away from the index. With the base held still, positive curl moved the tip farther away. The old endpoint checks missed that anatomical error.



Curl-only check (MCP 30 degrees, IP 27 degrees, base fixed): the old tip moved 19.4 mm away from the index; the revised tip moves 29.3 mm toward it.

What changed

Corrected the thumb frame by 180 degrees around its length; replaced the base rotation axis; centred the stationary fork behind the sweep; rerouted its control-cord approach. Curl stays palmar. The pad and nail colours mark their original CAD surfaces, not a moved contact label.

The base pin now has a 0-125 degree engineering sweep. This angle is not a human joint range. The single oblique pivot approximates coupled thumb opposition with the existing two thumb actuators; its revised mount is a physical design change.

Research basis

Li & Tang (2007), *Coordination of thumb joints during opposition*, found coupled CMC flexion/pronation and coordination with MCP flexion. PubMed / DOI 10.1016/j.jbiomech.2006.02.019.

In Vivo 3-Dimensional Kinematics of Thumb Carpometacarpal Joint During Thumb Opposition (2018) found abduction, internal rotation and flexion during opposition in nine volunteers. PubMed / DOI 10.1016/j.jhsa.2017.07.028. These studies inform the movement pattern; they do not validate this robot.

What has actually been checked

104 sampled body/cable configurations: **PASS**. Cover clearance: **PASS**. Three fixed-cube approaches, 81 configurations each: **PASS** for sampled rigid intrusion. These are geometric checks, with 0.02 mm body/cable and 0.1 mm cube-intrusion tolerances.

The 12 mm cube remains fixed during each approach. Heights -2, 0 and +2 mm meet the chosen pad-facing contact limits; -4 and +4 mm do not. Contact centres may shift on the face. Limits: 0.3 mm face-distance error, 15 degrees normal error and 1.5 mm edge margin. These are engineering screening choices, not proof of stable grasping.

Review against your seven requirements

Requirement	Status	Evidence and remaining work
1. Preserve V14	PASS	Same 75% scale, segment lengths, thumb-base location, seven pin joints and four base motors. Index poses are unchanged. Thumb frames and mount were revised as explicitly requested.
2. Look like a hand	PASS	The articulated V14 body shapes and proportions remain. The rendered meshes show the actual revised geometry; this is a CAD comparison.
3. Function like a hand	UNVERIFIED	Direction and pad-contact checks pass for the stated candidates. One oblique pin approximates opposition. Human-equivalent dexterity and real grasps have not been established.
4. Assembly and routing	UNVERIFIED	Nominal body/cable clearance: PASS; covers: PASS. Axial curl entry remains geometrically decoupled. Cleaning, threading, tensioning, feed liners and physical fit need testing.
5. Physical failures	UNVERIFIED	Retains R2 reinforced pin features and relieved Agilus pads. This revision does not establish resin strength, fatigue life or actual pad softness.
6. Cost and speed	UNVERIFIED	Same motor and assembled part counts. V14 / R2 / R3 CAD volume: 22.85 / 27.03 / 27.38 cm3. R3: +19.9% vs V14; +1.3% vs R2. Support, purge, time, labour and purchased cost are unmeasured.
7. Honest evidence	PASS	This is a kinematic/CAD review with stated tolerances and failures. No dynamics, force, friction or physical performance claim; no new printer-ready release.

Next physical check

Use a revised thumb-joint test assembly before committing to a full hand print: check palmar curl with the base held still, sweep opposition without unwanted curl, inspect retained tendon travel, then measure pad indentation and pinch/slip. The R2 print package does not contain this revised thumb mount.